

EBN111 Practical 2 Laboratory Guide

(To be studied for the practical 2 test)

2013

This guide serves as an introduction to the components, laboratory equipment and experimental methods used for building and testing electric circuits in the laboratory.

NB:

- Please read through this document before the practical.
- The practical test will cover the work in this document only.

Resistors

Introduction

A resistor is a passive electronic component (does not produce energy) that is used to implement an electrical resistance in a circuit. Resistors come in different shapes, sizes and materials. The resistors used in this practical are carbon film resistors.

Colour codes

Each carbon film resistor (such as those in your component boxes) has a specific resistance that has been set by the manufacturer. The value of this resistance is indicated by four coloured stripes on the resistor.

The first three stripes specify the resistance of the resistor (Table 1). The first two stripes form the first two digits of the resistance value, this 2-digit value is then multiplied by the value of the 3rd stripe. The result is the resistance of the resistor.

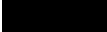
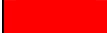
Colour	Colour name	1 st stripe (1 st digit)	2 nd stripe (2 nd digit)	3 rd stripe (multiplier)
	Black	0	0	x1
	Brown	1	1	x10
	Red	2	2	x100
	Orange	3	3	x1000
	Yellow	4	4	x10000
	Green	5	5	x100000
	Blue	6	6	x1000000
	Violet	7	7	-
	Grey	8	8	-
	White	9	9	-
	Gold	-	-	x0.1
	Silver	-	-	x0.01

Table 1. Resistor colour codes for the first three stripes. **Not to be memorised.**

The fourth stripe specifies the tolerance (Table 2). A tolerance of 5% (gold stripe) means that the actual resistance of the resistor can be any value within 5% of the value specified by the first three stripes. This deviation occurs because the manufacturing process is not perfect.

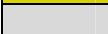
Colour	Colour name	4 th stripe (Tolerance)
	Gold	5%
	Silver	10%

Table 2. Resistor colour codes for the fourth stripe. **Not to be memorised.**

Figure 1 shows some examples. Note that a decimal point is often replaced by the SI prefix letter (eg. 2.7 k Ω can be written as 2K7 Ω).

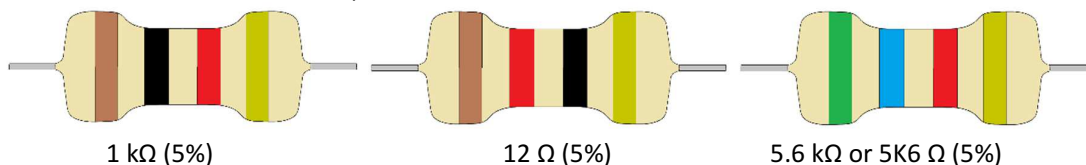


Figure 1. Examples of resistor colour codes.

Resistors are NOT polarised. This means that it does not matter which way you insert the resistor into your circuit.

Probes & connectors

Crocodile clips, banana plugs & BNC plugs

- Crocodile clips (Figure 2) can be attached to a component lead, wire, or another crocodile clip for making electrical connections between the laboratory instruments and your circuit.
- Banana plugs (Figure 3) connect to the power supply and multimeter.
- BNC plugs (Figure 4) connect to the function generator and oscilloscope.

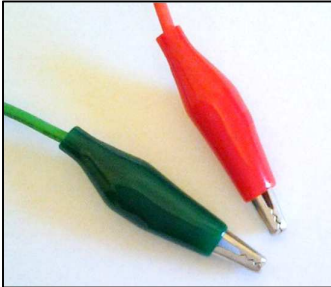


Figure 2. Crocodile clips.

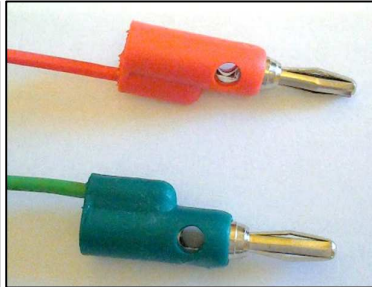


Figure 3. Banana plugs.



Figure 4. BNC plug.

High impedance probe

This probe (Figure 5) is used for monitoring high-frequency voltage signals on the oscilloscope. It is not necessary to use this probe during the EBN111 practicals.



Figure 5. High impedance, passive oscilloscope probe.

Protoboards

Introduction

Protoboards (or breadboards) are used to build prototype circuits. Component leads and wires are pushed into the holes in the protoboard to make electrical connections. You will build all your circuits on your protoboard for the EBN111 practicals.

Internal connections

The holes are internally connected as shown in Figure 6 by the thick lines. It may be useful to think of each line as a node in your circuit. Note that the long connections at the top and bottom may differ with other protoboard models.

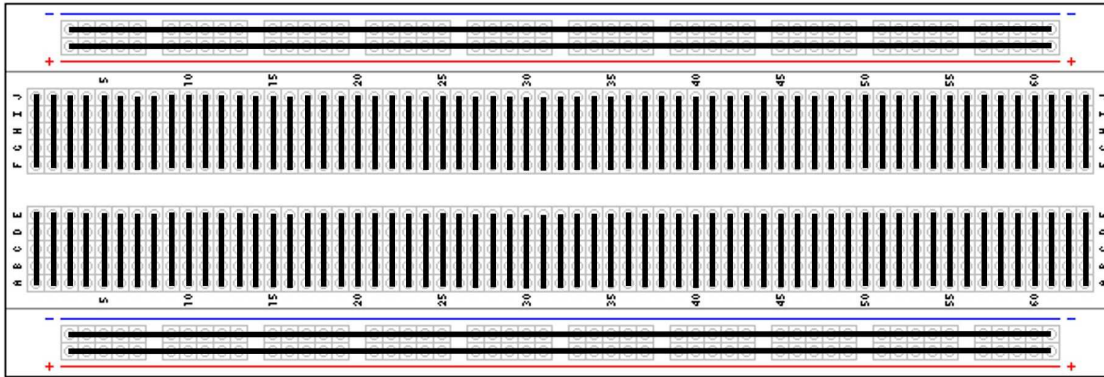


Figure 6. Internal connections of the protoboard. Thick lines indicate metallic connections.

Laboratory instruments

Power supply

A laboratory power supply (Figure 7) is used to supply electrical power to a circuit. Power supplies allow you to set the voltage that you need for your circuit and also allows you to limit the current (to prevent damage to the circuit if there are errors in the circuit). Do not use the power supply's earth connection ($\perp$) for your circuit's ground, use the **COM** connection.

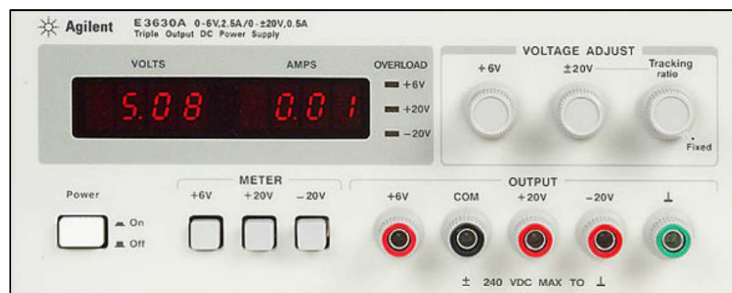


Figure 7. A laboratory power supply.

Multimeter

A digital multimeter (Figure 8) is an instrument capable of measuring voltage, current, resistance, and other parameters in a circuit.



Figure 8. A digital multimeter.

Function generator

A laboratory function generator is an instrument capable of generating voltages that vary with different waveforms. The function generator also allows you to set the amplitude and frequency of the generated signal. The function generators in the CAEC lab are built into the oscilloscopes (Figure 9).

Oscilloscope

An oscilloscope (Figure 9) is used to monitor varying voltages in a circuit. It plots a continuous graph with time on the x-axis and the measured voltage on the y-axis. Oscilloscopes also have the capability of measuring the frequency, amplitude and rise/fall time of voltage signals.

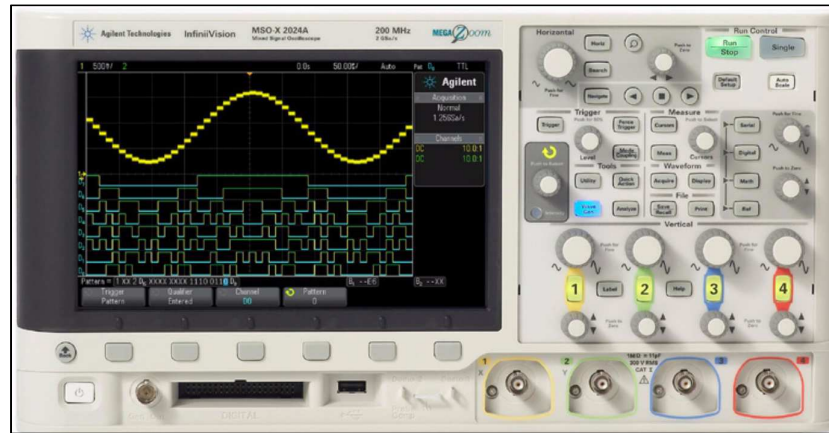


Figure 9. An oscilloscope with a built-in function generator.

Measurements

Voltage measurement

Remember that you measure voltage **across** a component, therefore you connect the multimeter probes across the component of interest. For example, measuring the voltage across a resistor is shown in Figure 10. Note that **the multimeter is connected in parallel** with the resistor.

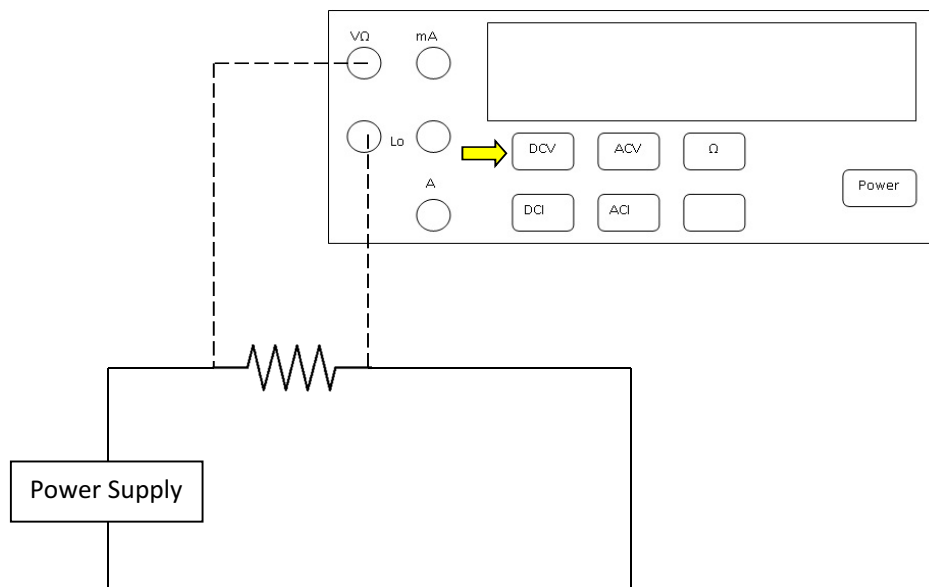


Figure 10. Measuring DC voltage using the multimeter.

Current measurement

Remember that you measure the current **"through"** a component. You therefore have to connect the circuit so that the current you wish to measure flows through the multimeter. The current measured through a resistor is shown in Figure 11. Note that **the multimeter is connected in series** with the resistor.

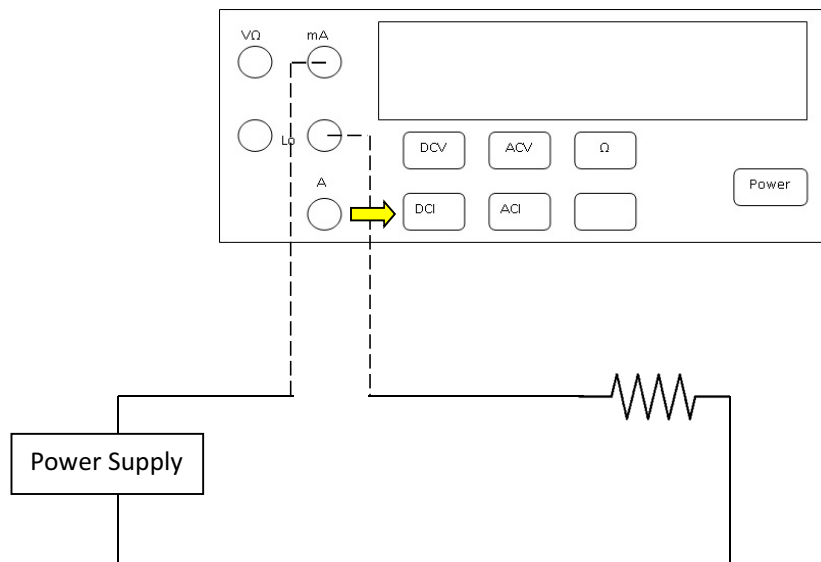


Figure 11. Measuring DC current using the multimeter.

Resistance measurement

The multimeter measures resistance by applying a constant current through the resistor (Figure 12). The multimeter then measures the voltage across the component and calculates the resistance using Ohm's law. **This measurement only works when the component is disconnected from the circuit.**

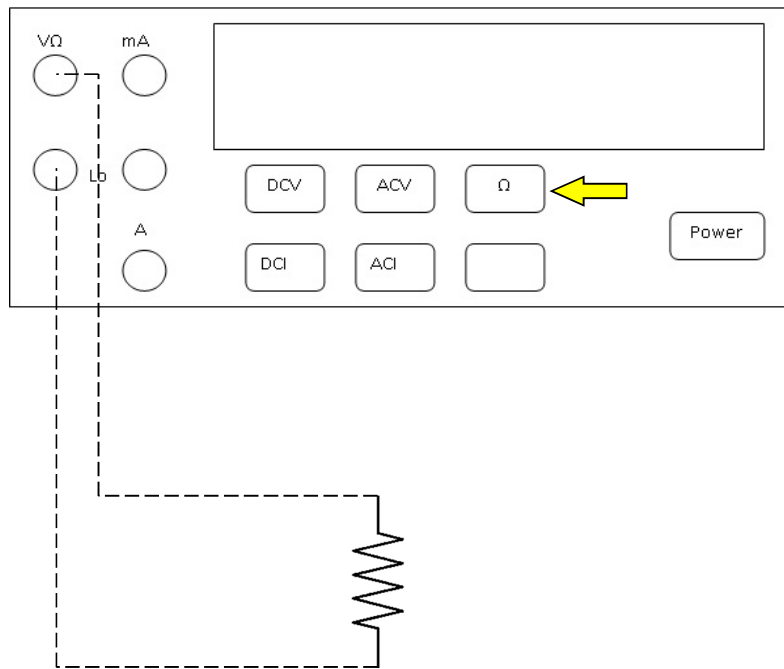


Figure 12. Measuring resistance using a multimeter.